

ABHIJITH SIVAPRASADAN

M.Sc. Sustainable Energy Engineering (KTH, 2026) | Heat Pumps | Refrigeration Cycles | Testing & Control
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PROFILE

Thermal and mechanical engineer completing an M.Sc. in Sustainable Energy Engineering at KTH, focused on heat pumps, refrigeration-cycle fundamentals, heat transfer, testing and control-oriented analysis. Built R&D experience at Siemens Energy with high-temperature thermal-fluid rig modelling, NI-DAQ/LabVIEW measurement-chain commissioning and simulation-to-test validation thinking. Strong foundation in thermodynamics, vapor-compression cycles, COP/SCOP reasoning, heat-exchanger behaviour and Python/MATLAB analysis. Seeking to contribute to Weheat's Heat Loop & Control team by linking models, test data, component reasoning and robust production-focused decisions.

EXPERIENCE

Test Engineer / Master Thesis Student

Siemens Energy AB, Fluid Dynamic Lab, Finspång

Apr 2025 - Nov 2025

Thesis: Numerical Investigation of Steady-State Thermo-Fluid Performance of a Reducer for a High-Temperature Dynamic Pressure Sensor Calibration Rig. Supervisor: Prof. Jens Fridh, KTH.

- Developed steady-state compressible CFD and conjugate heat-transfer models in ANSYS Fluent to quantify pressure loss and thermal delivery in high-temperature gas-turbine test-rig hardware.
- Established repeatable model-validation workflow: boundary-condition definition, mesh/near-wall checks, sensitivity studies, KPI definitions and interpretation of assumptions/limitations.
- Commissioned and verified NI-DAQ measurement chains with thermocouples and static/dynamic pressure sensors; refined LabVIEW acquisition workflows and supported preliminary high-temperature rig testing.
- Linked simulations to measurable rig quantities and documented next-step experimental verification needs - transferable to prototype heat-pump loop testing and control validation.

Energy Efficiency Intern

Alleima AB, Sandviken (Remote)

Dec 2024 - Jun 2025

- Built ISO 50001/EED-aligned energy performance mapping with KPI/EnPI definitions, baselines, load drivers and normalisation logic for comparing operating states.
- Assessed metering/data readiness and developed workflows to identify deviations in operational energy data, relevant to diagnosing real-world heating and utility systems.

Backend Engineer

QBurnt, Thiruvananthapuram, India

Aug 2021 - Jul 2023

- Developed APIs, data workflows and automated test pipelines using Python, TypeScript, Postman, Git and CI-oriented practices; built habits in debugging, reproducibility and maintainable delivery.
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RELEVANT PROJECTS

Residential Heating System Modelling and Heat Pump Comparison

KTH Course Project (IDA ICE, SAM, techno-economic analysis)

2025

- Compared district heating, PV with boiler, PV with heat pump, and solar thermal with TES using hourly demand and irradiance data; evaluated CAPEX, OPEX, LCOE and payback.

Heating Dispatch Optimisation - District Heating System

KTH Course Project (Python, PuLP, MILP)

2025

- Implemented multi-asset heat-supply dispatch optimisation with operational constraints, heat-pump/CHP operation, electricity-price coupling, CO2 cost, storage and KPI reporting.

Day-Ahead Building Heating Demand Forecasting

KTH Course Project (Python, scikit-learn, time-series)

2025

- Built heating-demand forecasting workflows using weather and temporal features; compared baseline and neural-network models with feature engineering, error analysis and model evaluation.

Thermal Energy Storage for District Cooling Peak Shaving

KTH Course Project (Python)

2024

- Simulated TES control logic to cap peak cooling demand and compared ice and phase-change material options on efficiency, cost and volume trade-offs.

EDUCATION

M.Sc. in Sustainable Energy Engineering (expected 2026)

KTH Royal Institute of Technology, Stockholm

2023 - 2026

Relevant coursework: Applied Heat and Power Technology, Energy in the Built Environment, Numerical Heat Transfer, Modelling of Energy Systems, Practical Optimization of Energy Networks, AI Applications in Sustainable Energy Engineering and Energy Management.

B.Tech in Mechanical Engineering

APJ Abdul Kalam Technological University, Kerala, India

2017 - 2021

Relevant foundation: thermodynamics, heat and mass transfer, fluid mechanics, thermal engineering, mechanical systems and CAD/CAE.

TECHNICAL SKILLS

Heat pumps / refrigeration:	Vapor-compression cycle fundamentals; COP/SCOP reasoning; evaporator/condenser behaviour; superheat/subcooling concepts; heat-exchanger fundamentals; hydronic/heating systems; R290 awareness and active ramp-up.
Controls / loop operation:	Control-oriented modelling; PI/PD/PID concepts; EXV and superheat-control fundamentals; operating-regime reasoning; residual interpretation; test-data-driven tuning mindset.
Thermal-fluid modelling:	Thermodynamics, heat transfer, fluid dynamics, pressure-loss and thermal-delivery analysis; ANSYS Fluent/CHT, k-omega SST, mesh/near-wall checks, sensitivity studies.
Testing / validation:	NI-DAQ, NI MAX, LabVIEW, thermocouples, static/dynamic pressure sensors, signal verification, measurement-chain commissioning, validation planning and documentation.
Programming / tools:	Python (pandas, NumPy, SciPy, matplotlib, scikit-learn, PuLP), MATLAB scripting basics, Git, Docker, automated tests, IDA ICE, HOMER Pro, SAM, Siemens NX, SolidWorks, ANSYS Meshing/SpaceClaim.

CERTIFICATIONS AND LANGUAGES

Certifications: Applied CFD (Siemens) | SSG Entre / Industrial Safety | IBM Python for Data Science | Digital Manufacturing and Design Technology | Schneider Electric sustainability course

Languages: English - professional proficiency | Swedish - beginner, actively studying | Malayalam - native